

Kiran Girishkumar

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Professional Summary

MSc Information and Communication Technology student at FAU Erlangen-Nürnberg, specializing in Physical Design and the RTL-to-GDSII flow, with a B.Tech in Electronics and Communication Engineering. Built a self-directed PD portfolio in OpenLane/OpenROAD (open-source counterparts to Cadence Innovus / Synopsys ICC2) covering floorplanning, timing closure, PDN design, and DRC/LVS sign-off. Comfortable with STA, Tcl scripting, and PPA optimization in a Linux environment. Seeking a Werkstudent or master-thesis position in Physical Design.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg

Master of Science in Information and Communication Technology — Focus: Embedded Systems

Nuremberg, Germany

October 2025 – Present

Sree Chitra Thirunal College of Engineering

Bachelor of Technology in Electronics and Communication Engineering

Trivandrum, India

November 2020 – June 2024

Technical Skills

Languages: Verilog, Tcl, C, Embedded C, Python.

EDA Tools: OpenROAD, OpenROAD-flow-scripts (ORFS), OpenLane, Yosys, OpenSTA, KLayout, iVerilog, GTKWave, Cadence Virtuoso, Xilinx Vivado.

Physical Design: Floorplanning, CTS, Routing, Static Timing Analysis (STA), Multi-Corner Multi-Mode (MCMM) Analysis, DRC/LVS Sign-off, PPA Optimization, Parasitic Extraction (RCX), IR Drop/EMIR Analysis, Power Distribution Network (PDN).

Environment: Ubuntu 24.04 (ARM64), FPGA (Xilinx), RISC-V Architecture.

Experience

Vikram Sarabhai Space Centre (VSSC)

Project Intern

Trivandrum, India

October 2023 – April 2024

- Built an 868 MHz ASK backscatter link for battery-free IoT sensor nodes, sustaining 100 kbps over a 3.5 m range at 16 dBm reader transmit power within a $\sim 100 \mu\text{W}$ node power budget.
- Implemented FM0 line-coding and modulation firmware in Embedded C on PIC16F1719 MCU (MPLAB X IDE), driving the backscatter switch at the target symbol rate.
- Designed and verified the real-time FPGA receiver chain in LabVIEW FPGA digital filtering, envelope detection and FM0 decoding and validated end-to-end link performance on hardware.

Projects

STA Timing-Closure AI Agent | OpenSTA, Python, Streamlit, Docker, Fly.io

May 2026 - August 2026

- Built an LLM-driven agent that parses OpenSTA timing reports, diagnoses setup/hold violations, and applies iterative ECO-style fixes (cell resizing, buffering) with automatic keep/revert decisions based on WNS/TNS deltas.
- Closed timing autonomously on a PicoRV32 netlist: WNS $-0.38 \rightarrow 0.00$ ns in 8 moves at a 3.8 ns period, and WNS $-0.18 \rightarrow 0.00$ / TNS $-0.52 \rightarrow 0.00$ ns in 3 moves at 4.0 ns, with negligible area cost.
- Compiled OpenSTA from source for x86_64 and packaged it with the app in a multi-stage Docker image; deployed publicly on Fly.io — live at sta-agent-docker.fly.dev | github.com/kiran-kichu/sta-timing-agent.

Multi-Corner Multi-Mode (MCMM) Timing Sign-off | OpenROAD, OpenSTA, Sky130 PDK

March 2026 - July 2026

- Performed MCMM STA across PVT corners (SS/TT/FF) and functional/test modes, closing setup at the slow corner and hold at the fast corner.
- Automated multi-corner SDC/constraint management with Tcl, analyzing all corner-mode scenarios in a single run.

Hierarchical SoC Integration & Power Grid Design | OpenROAD, Tcl

April 2026 - June 2026

- Integrated CPU and FIFO macros into a unified SoC, designing a multi-layer Power Distribution Network (PDN).
- Closed timing at 66.7 MHz (0 ns WNS, 0 setup/hold violations), with a worst-case IR drop of 11 μV with strategic macro placement and power strap sizing, with Tcl-automated PDN parameter sweeps.

AI Accelerator: 8×8 Systolic-Array MAC (RTL-to-GDSII) | OpenROAD, Nangate45

April 2026 - May 2026

- Designed an 8×8 output-stationary systolic-array matrix-multiply engine (64 MAC PEs) from scratch in parameterizable Verilog — the compute core of AI/ML accelerators.
- Closed timing at 625 MHz with zero negative slack post-route and 0 DRC violations, delivering ~ 40 GMAC/s (80 GOP/s) on a 34k-cell, 0.062 mm^2 layout.

Sign-off Physical Design of PicoRV32 RISC-V Core | OpenROAD-flow-scripts, Sky130 PDK

February 2026 - April 2026

- Drove the full RTL-to-GDSII flow for a PicoRV32 RV32I core on SkyWater sky130, closing timing at **238 MHz** with **zero setup/hold violations, zero DRC and zero antenna violations** at post-route sign-off.
- Ran a 15-point clock-period sweep ($10.0 \rightarrow 3.8$ ns), improving the post-route critical path **29.5%** ($5.78 \rightarrow 4.07$ ns) through constraint-driven synthesis and timing repair, at a cost of 2.8% cell area and no die-size increase.
- Verified power integrity with post-route parasitic extraction (RCX) and IR-drop analysis: worst-case IR drop 549 μV , 0.03% of the 1.8 V supply.

Achievements

GATE 2025: Secured All India Rank 9109 in Electronics and Communication Engineering.

VLSI Training: Certified in professional HDL and FPGA design via the SURE Trust program.

Languages: English (C1 Proficiency), German (A2 Proficiency).